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Large-scale ML (CS 260D) released: November 5

# Homework 3

Submit to Gradescope (in groups of 4) due: November 12, 12pm

**Instructions:**

1. Please create *one* submission per group and tag your other group members in Gradescope.
2. Please ensure you follow the format specified in the Gradescope online assignment. We will be using automatic grading and incorrect formats will not receive any credit.
3. There will be no partial credit for this assignment.

## Question 1: Data Selection for Supervised Learning (20 points)

In this question, each data point has two features. Each plot represents a specific data distribution and the decision boundary of the trained **linear classifier** is being trained to perform binary classification between **circles** and **crosses**.

In all the subsequent questions, assume the data-selection algorithm selects **exactly 4 examples.**

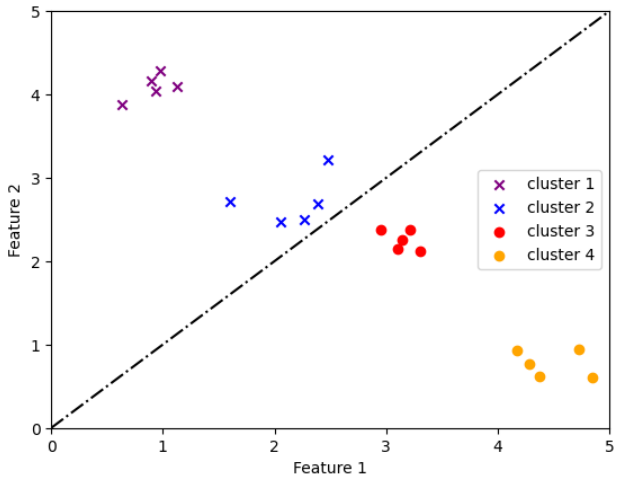



Figure 1: Data Selection (part 1 and 2)

1. **(3 points)** Refer to Fig. 1. How many clusters do the data points selected by **Forgetting Score Algorithm** belong to?

A. 1

B. 2

C. 3

D. 4

Forgetting Score Algorithm forgets unforgettable points, so it keeps data close to the decision boundary

2. **(2 points)** Refer to Fig. 1. Which clusters do the data points selected by **Forgetting Score Algorithm** belong to?

A. Cluster 1

B. Cluster 2

C. Cluster 3

D. Cluster 4

3. **(3 points)** Refer to Fig. 1. How many clusters do the data points selected by **CRAIG** belong to?

A. 1

B. 2

C. 3

D. 4

CRAIG uses GreeDI and finds exemplars then weights them during gradient descent based on the size of the cluster

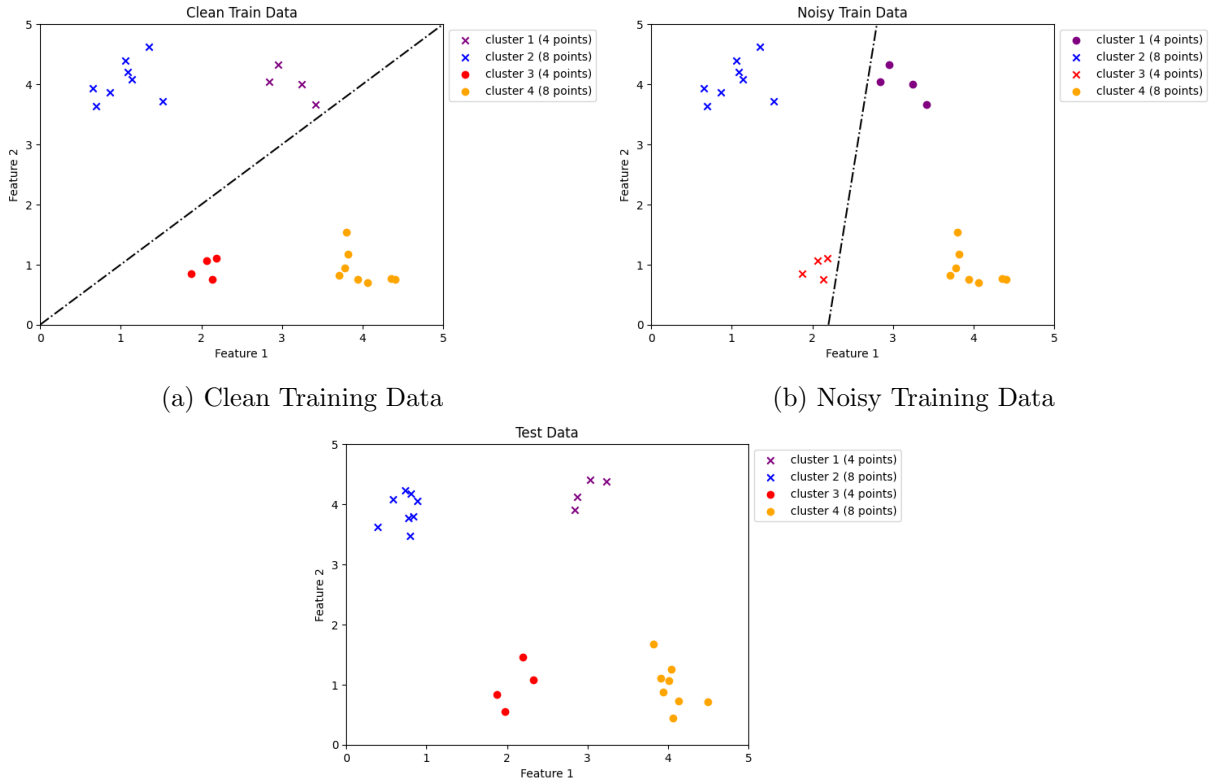
4. **(2 points)** Refer to Fig. 1. Which clusters do the data points selected by **CRAIG** belong to?

A. Cluster 1

B. Cluster 2

C. Cluster 3

D. Cluster 4




(a) Clean Training Data

(b) Noisy Training Data

(c) Test Data

Figure 2: Data Selection in the presence of Noisy Labels (part 3 and 4)

For the following questions, refer to Fig. 2. Here Fig. 2a shows the clean training data and Fig. 2b shows the noisy training data available to the data-selection algorithms in this section. The dotted line indicates the decision boundary of the trained model. In the noisy training data, the points in Cluster 1 and Cluster 3 have their labels flipped as shown in Fig. 2b. Fig. 2c shows the test data.

*Note: Assume that the models trained on the selected subsets and the full noisy training data achieve* 100% **training** `accuracy`. (*This means assume that the trained model perfectly classifies the selected subset from the noisy training data. The selected subset depends on the selection algorithm we use.*)

5. **(3 points)** Is *training on the subset selected by the* ***Forgetting Score Algorithm*** likely to achieve the same test accuracy as *training on the* ***full noisy training data***?

*Note: For simplicity, assume that the decision boundary shown in 2b is used for data subset selection with the Forgetting Score Algorithm.*

A. Yes

B. No

6. **(3 points)** Is *training on the subset selected by the* ***CRAIG*** likely to achieve similar test accuracy to *training on the* ***full noisy training data***?

A. Yes

B. No

7. **(4 points)** Training on the subset selected by which of the following algorithms is likely to achieve a higher **test** accuracy?

A. Forgetting Score Algorithm

B. CRAIG

Exemplar based clustering should be more robust to noisy training data then keeping points close to the decision boundary

## Question 2: Data Selection for Fine-tuning LLMs (15 points)

We are given 14 examples. Each example $x_i$ is a 2-dimensional vector: its loss at two training checkpoints of the proxy model, $x_i = [\text{loss}_{\text{ckpt1}}, \text{loss}_{\text{ckpt2}}]$.

$$x_1 = [8.0, 2.5] \qquad x_8 = [7.5, 8.0]$$
$$x_2 = [2.5, 2.5] \qquad x_9 = [8.5, 2.0]$$
$$x_3 = [7.5, 2.0] \qquad x_{10} = [2.5, 8.0]$$
$$x_4 = [3.0, 2.0] \qquad x_{11} = [2.0, 3.0]$$
$$x_5 = [8.5, 2.5] \qquad x_{12} = [7.5, 2.5]$$
$$x_6 = [8.0, 3.0] \qquad x_{13} = [8.5, 7.5]$$
$$x_7 = [2.0, 2.0] \qquad x_{14} = [8.0, 2.0]$$

**Clusters.** The examples form well-separated clusters: any two examples in the same cluster are at Euclidean distance at most 1.5, and any two examples in different clusters are at distance more than 4. A cluster may contain a single example.

**S2L.** S2L selects a subset $S$ of size $B = 8$ that has an **equal** number of examples from every cluster when possible (not a number proportional to the cluster size):

1. Sort the $K$ clusters by size, from smallest to largest: $C_1, C_2, \ldots, C_K$.
2. Start with $S = \emptyset$. For $k = 1, 2, \ldots, K$:
   (a) Compute $R_k = \left\lfloor \frac{B - |S|}{K - k + 1} \right\rfloor$: the number of examples still needed divided by the number of clusters left, rounded down.
   (b) If $|C_k| \leq R_k$, add all examples of $C_k$ to $S$. Otherwise, add $R_k$ examples of $C_k$ to $S$.

S2L picks the $R_k$ examples at random. In this question, instead pick the $R_k$ examples with the **smallest indices**. For example, picking 2 examples from a cluster $\{x_3, x_5, x_{10}, x_{12}\}$ gives $x_3$ and $x_5$.

1. (3 points) How many clusters $K$ do these 14 examples form?
   A. 2
   B. 3
   C. 4
   D. 5
2. (3 points) How many examples does S2L select from the **largest** cluster?
   A. 2
   B. 3

C. 4

D. 5

1,2,3,4,5,8,10,13

3. (3 points) What is the final subset $S$ selected by S2L? Enter the indices in **increasing order**, separated by commas without spaces (e.g., enter `1,2,13` for $S = \{x_1, x_2, x_{13}\}$).

4. (3 points) Another method is **proportional sampling**. It is the same as S2L (same clusters, same order in step 1, same step 2(b), and the same smallest-indices rule), except that step 2(a) becomes:

$$R_k = \text{round}\left((B - |S|) \cdot \frac{|C_k|}{|C_k| + |C_{k+1}| + \cdots + |C_K|}\right),$$

that is, the number of examples still needed times the share of $C_k$ among the clusters left, rounded to the nearest whole number (with .5 rounded up).

Start from $B = 8$ and increase $B$ by 1 at a time. What is the **smallest** $B$ for which S2L and proportional sampling select the same number of examples from the largest cluster?

13

5. (3 points) Compared with proportional sampling, which statements about S2L are true? Select all that apply.

A. S2L gives small clusters a larger share of the selected subset.

B. S2L keeps the cluster distribution of the selected subset the same as in the full data.

C. S2L tends to improve performance on small groups of the data, such as rare types of examples.

D. When $B \geq K$, S2L selects at least one example from every cluster.

## Question 3 (20 points): Spurious Correlations

Introducing SpuCoMNIST: In this homework question, we delve into SpuCoMNIST, a dataset designed for colored digit recognition. SpuCoMNIST is an extension of the classic MNIST dataset, where additional colored backgrounds have been introduced as spurious features. This task involves recognizing digits while dealing with these added complexities. With SpuCoMNIST, one can explore various factors, such as the number of classes, group sizes, and the ability to control the magnitude and variance of the spurious features. Furthermore, it offers the opportunity to experiment with label noise and noise affecting the core features.

Here we initialize SpuCoMNIST with 5 classes, each formed by merging two MNIST digits: class 0 contains digits $\{0, 1\}$, class 1 contains digits $\{2, 3\}$, class 2 contains digits $\{4, 5\}$, class 3 contains digits $\{6, 7\}$, and class 4 contains digits $\{8, 9\}$, as specified here: https://colab.research.google.com/drive/1ZVrPhi8xJ4F3vy4tpPlIumOcL7vmiV31?usp=sharing, and investigate the effect of different parameters controlling spurious correlation on the accuracies of various groups.

A group $(y, s)$ is indexed by the class label $y \in \{0, \ldots, 4\}$ (not the original MNIST digit) and the background-color label $s \in \{0, \ldots, 4\}$. For example, group $(2, 3)$ contains digits 4 or 5 (class 2) on the background color associated with class 3.

**Experimental protocol.** For Parts 1 and 2, use the default settings: correlation strength 0.995, `MAGNITUDE_LARGE`, label noise 0.001, seed 0, and three ERM epochs. For every parameter setting in Parts 3 and 4, reset the random seed to 0 and rerun dataset creation, model initialization, training, and evaluation from scratch. Do not continue training a model from a previous setting. Keep all other settings unchanged, and apply the chosen difficulty to the training, validation, and test datasets. The reference experiments use `spuco==2.0.3`.

1. **Evaluate ERM (5 points).** Run the evaluation code. Is the accuracy of examples in group $(2, 3)$ higher than, lower than, or equal to that of examples in group $(2, 2)$?

   A. Higher

   B. Lower

   C. Equal

2. **Evaluate ERM (5 points).** Continuing from Part 1, why does this difference arise?

   A. Group $(2, 3)$ is mislabeled.

   B. The digits in group $(2, 3)$ are inherently harder to recognize than those in group $(2, 2)$.

   C. Group $(2, 3)$ is a minority in class 2; group $(2, 2)$ is the majority in class 2.

3. **Spurious correlation strength (5 points).** Keep the spurious feature difficulty at `MAGNITUDE_LARGE`. Now vary the spurious correlation strength from (1) 0.9 to (2) 0.965 to (3) 0.995. Consider the change from (1) to (2), followed by the change from (2) to (3). What is the effect of increasing the spurious correlation strength on the accuracy of group $(2, 3)$?

   A. Increases then decreases

B. Decreases then increases

C. Increases then stays the same

D. Decreases then stays the same

E. Increases

F. Decreases

G. Stays the same

4. **Spurious feature difficulty (5 points).** Set the spurious correlation strength to 0.995. Now vary the spurious feature difficulty from (1) `MAGNITUDE_LARGE` to (2) `MAGNITUDE_MEDIUM` to (3) `MAGNITUDE_SMALL`. Consider the change from (1) to (2), followed by the change from (2) to (3). What is the effect on the worst-group accuracy?

   A. Increases then decreases

   B. Decreases then increases

   C. Increases then stays the same

   D. Decreases then stays the same

   E. Increases

   F. Decreases

   G. Stays the same

## Question 4: Poisoning Attacks and EPIC (14 points)

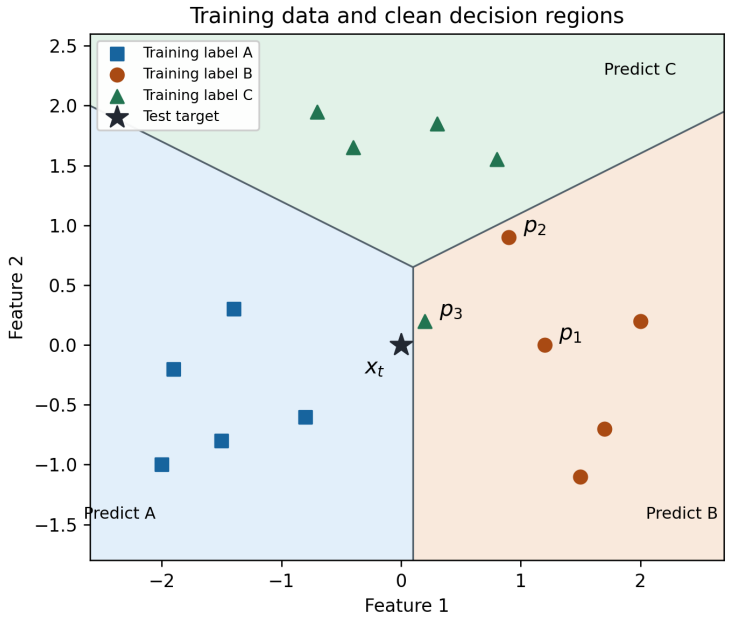



Figure 3: The markers show the training labels, and the shaded regions show the clean linear classifier's predictions. The star marks the held-out test point $x_t = (0, 0)$, whose true class is A. The clean classifier correctly predicts A for $x_t$.

### Q1. Targeted clean-label poisoning (5 points)

An attacker wants the classifier to predict class B for $x_t$ after retraining. The attacker can perturb exactly one of the three training points below. Each coordinate may change by at most 1, so $\|\delta\|_\infty \leq 1$. The point's training label must stay fixed.

| Candidate | Original coordinates | Training label |
|---|---|---|
| $p_1$ | $(1.2, 0)$ | B |
| $p_2$ | $(0.9, 0.9)$ | B |
| $p_3$ | $(0.2, 0.2)$ | C |

**Fill in the blanks.** To best carry out the targeted clean-label poisoning attack within the perturbation budget, change $p$___ to (_____, _____) and use training label _____.

### Q2. Poisoning under a tighter budget (4 points)

Use the same target $x_t = (0, 0)$ and the same class-B candidates $p_1 = (1.2, 0)$ and $p_2 = (0.9, 0.9)$. For each budget, start from the **original** point locations in Figure 3.

The attacker can change only one training point and must keep its label fixed. Choose the point and perturbation that minimize the Euclidean distance between the perturbed point and $x_t$, subject to $\|\delta\|_\infty \leq \varepsilon$.

(a) **(2 points)** For $\varepsilon = 0.10$, which point should the attacker change, and where should it end up?

(b) **(2 points)** Answer the same question for $\varepsilon = 0.50$.

**Q3. Detecting poisons with EPIC (5 points)**

EPIC examines the following loss gradients for six training examples, all labeled B.

| **Example** | $b_1$ | $b_2$ | $b_3$ | $b_4$ | $b_5$ | $b_6$ |
|---|---|---|---|---|---|---|
| **Gradient** | $(0, 0)$ | $(0, 1)$ | $(1, 0)$ | $(3, 0)$ | $(3, 1)$ | $(1, 4)$ |

EPIC selects $b_1$, $b_4$, and $b_6$ as medoids and assigns each example to its nearest medoid by Euclidean distance in gradient space. EPIC then drops any selected medoid whose assigned cluster contains only itself.

**Which statement correctly identifies what EPIC drops and why? Select one.**

A. Drop $b_4$. A cluster with two examples is small enough to count as a singleton.

B. Drop $b_6$. Its cluster contains only itself, which proves its training label was changed.

C. Drop $b_6$. Its cluster contains only itself, and effective clean-label poisons can have gradients that differ from those of other examples with the same label.

D. Drop no examples. Because all six examples have the same label, they belong to a single cluster.

## Question 5: Neural Network Pruning (10 points)

Assume we want to prune a network with 1 million parameters to 80% sparsity (i.e. we want *at most* 200K parameters in the pruned model), using iterative magnitude pruning with rewinding to epoch 5. The original training pipeline has 200 epochs and we prune 20% of the remaining weights at every pruning iteration.

*Note:* after we get the desired sparsity, we rewind to epoch 5 and train the model to convergence.

1. **(3 points)** How many times do we need to rewind to get the final pruned model?

   A. 1
   B. 2
   C. 4
   D. 8
   E. 10

   \log_{0.8}0.2 ~ 7.212
   So you need 8
   0.8^8 ~ 0.16
   0.8^7 ~ 0.21

2. **(4 points)** How many epochs of training do we need in total (including the original training pipeline) to get the final pruned (and trained) model?

   A. 980
   B. 1600
   C. 1760
   D. 1800

   200 * .212 = 42.4
   200*7 + 200*.212 = 1442.4
   200*8 = 1600
   200 + 195*7 = 1565

3. **(3 points)** How much theoretical speedup (measured by FLOPs before pruning divided by FLOPs after pruning) is possible? Select all that applies.

   Hint: Consider the VGG example mentioned in class.

   A. No speed up.
   B. 2× speed up.
   C. 5× speed up.
   D. 8× speed up.

   1,000,000/200,000 = 5
   Compression Ratio: total_params/non_zero_params
   Theoretical Speedup: total_flops/non_zeroflops